

Inverter (LT3462A) — Working Document

Digital Audio Player · Rev 0

Author: Alexander Loker — design decisions, component selection, analysis and verification.
Drafted by: Claude (Anthropic) — structure and prose from the author's direction and source data.

1. Part Identification

Field	Value
Manufacturer / MPN	Analog Devices — LT3462AES6
Package	6-lead TSOT-23 (S6)
Function	2.7 MHz current-mode inverting converter, integrated Schottky; +3V9 → −3.9 V (LT3093 input)

2. Design Requirements

Requirement	Value
Input	+3V9 from LTC3440, 3.827 – 3.981 V (min 3.74 V during load step)
Output	−3.9 V, regulated
Load	LT3093 only
Output current	≈ 27 mA quiescent; ≈ 98 mA peak (0.6 V RMS max); 125 mA LT3093 limit
Topology	Dual inductor inverting (Ćuk)
Switching frequency	2.7 MHz fixed (2.0 – 3.5 MHz), no Burst Mode
Enable	Follows +3V9 (SDREF pull-up)

Constraints: Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	SW	L1, C1 (+)
2	GND	GND
3	FB	R1 / R2 midpoint, C4 to −3V9
4	SDREF	R2 to FB
5	D	C1 (−), L2
6	VIN	+3V9, C_IN

4. External Components

Ref	Value	Rating / package	MPN	AL-PN
L1, L2	22 μ H (or 10 μ H coupled)	Shielded, $I_{sat} \geq 0.35$ A, DCR $< 1 \Omega$	TBD	—
C1 (flying)	4.7 μ F	16 V X7R 0805 ($> V_{IN} - V_{OUT} = 7.8$ V)	Murata GRM21BR71C475KE51L	#1-071
C_IN	4.7 μ F	16 V X7R 0805	Murata GRM21BR71C475KE51L	#1-071
C_OUT	10 μ F (6.76 μ F eff. at 3.9 V)	10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
R1	309 $k\Omega$ 0.1%	0402	YAGEO RT0402BRD07309KL	—
R2	100 $k\Omega$ 0.1%	0603	YAGEO RT0603BRD07100KL	#1-066
C4	22 pF	50 V C0G 0402	Murata TBD	—

5. Equations and Results

Output voltage (R1 FB $\rightarrow$ V_OUT, R2 SDREF $\rightarrow$ FB)

$$V_{OUT} = -1.265 V \times \frac{R_1}{R_2} \Rightarrow R_1 = R_2 \times \frac{3.9}{1.265} = 308.3 k\Omega \rightarrow 309 k\Omega$$

$$V_{OUT} = -1.265 \times \frac{309}{100} = -3.909 V \quad (-3.816 \text{ to } -3.971 V \text{ over SDREF-FB } 1.235-1.285 V)$$

Divider current (SDREF can source 80 μ A)

$$I_{R2} = \frac{1.265 V}{100 k\Omega} = 12.7 \mu A$$

Duty cycle ($V_D = 0.8$ V Schottky, $V_{CE} = 0.27$ V switch)

$$D = \frac{|V_{OUT}| + V_D}{V_{IN} - V_{CE} + |V_{OUT}| + V_D} = \frac{4.709}{3.634 + 4.709} = 56.4\% \quad (57.6\% \text{ at } 3.74 V; \text{ max } 77\%)$$

Inductor ripple (each inductor, 22 μ H)

$$\Delta I_L = \frac{V_{IN} D}{f L} = \frac{3.904 \times 0.564}{2.7 MHz \times 22 \mu H} = 37 mA_{pp} \quad (50 mA_{pp} \text{ at } 2.0 MHz)$$

Inductor and switch current (98 mA peak output, $\eta = 0.65$ assumed)

$$I_{L1} = \frac{|V_{OUT}| I_{OUT}}{\eta V_{IN}} = \frac{3.909 \times 0.098}{0.65 \times 3.904} = 151 mA, \quad I_{L2} = I_{OUT} = 98 mA$$

$$I_{SW, pk} = I_{L1} + I_{L2} + \Delta I_L = 151 + 98 + 37 = 286 mA \quad (299 mA \text{ at } 2.0 MHz)$$

Output current	Switch peak	vs 300 mA min / 420 mA typ limit
27 mA (quiescent)	≈ 0.10 A	✓
69 mA (110 dB SPL peaks)	≈ 0.21 A	✓
98 mA (full scale, 23.8 Ω)	≈ 0.29 A	At minimum limit
125 mA (LT3093 limit)	≈ 0.36 A	Inverter limits first

Inrush (undamped upper bound; Schottky surge limit 1.5 A)

$$I_P \leq V_{IN} \sqrt{\frac{C_1}{L_1}} = 3.9 \times \sqrt{\frac{2 \mu F}{22 \mu H}} = 1.18 \text{ A} \quad (C_1 \approx 2 \mu F \text{ effective at } 7.8 \text{ V})$$

6. Important Values

Item	Value
Input range	2.5 – 16 V
Switch current limit	300 mA min / 420 mA typ (varies with duty cycle)
Switch V_{CESAT}	270 mV typ / 350 mV max at 250 mA
Schottky forward drop	800 mV typ / 1100 mV max at 250 mA
Maximum duty cycle	77% (LT3462A)
SDREF / reference	1.265 V (1.245 – 1.285 V), up to 80 μ A
FB bias current	15 nA typ / 50 nA max
Supply current	2.9 mA typ, not switching; < 10 μ A shutdown
Output	–3.909 V (–3.816 to –3.971 V)
Duty cycle	56.4% at 3.9 V in
Switch peak at 98 mA	≈ 0.29 A
θ_{JA}	150 $^{\circ}$ C/W

7. Open Problems

#	Problem
1	Switch peak ≈ 0.29 A at full-scale output vs 300 mA minimum limit — read the current limit vs duty cycle curve at 56%
2	Efficiency assumed 65% — confirm in LTspice; sets LTC3440 load and battery cost
3	Inductor MPN — shielded, 22 μ H, $I_{sat} \geq 0.35$ A, DCR < 1 Ω ; coupled 10 μ H alternative
4	C1 effective capacitance at 7.8 V (SimSurfing) — sets inrush
5	C4 value — datasheet uses 22 pF with 267 k Ω ; verify loop in LTspice
6	Soft-start (C _S from SDREF to V _{OUT}) — needed, or does the LTC3440 ramp suffice?
7	Confirm R1 MPN; θ_{JA} 150 $^{\circ}$ C/W — check dissipation at peak
8	Sims: start-up and inrush, load step 27 $\leftrightarrow$ 98 mA, ripple at 2.7 MHz into LT3093, efficiency vs load

8. Revision Log

Rev	Date	Change
0	2026-09-23	Initial issue: part selection, divider, duty cycle, switch current, inrush.